Tejas Malhotra

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Education

Masters in Information and Communication Technology 10/2025 – Present
Friedrich-Alexander University of Erlangen–Nuremberg Erlangen, Germany

Bachelor of Technology in Electronics and Communication Engineering 09/2020 – 10/2024
Manipal University Jaipur Jaipur, India

Professional Experience

Cloud Intern 01/2024 – 05/2024
Parity Infotech Solutions Bangalore, India
- Reduced deployment time by 40% by designing and maintaining automated CI/CD pipelines using GitHub Actions and GitLab CI/CD, enabling reliable, repeatable build, test, and release workflows across all development and staging environments.
- Improved system availability by 25% by collaborating on fault-tolerant AWS cloud infrastructure (EC2, VPC, RDS, S3) using fault-tolerant architecture patterns, ensuring service continuity and minimizing unplanned downtime across production workloads.
- Decreased data retrieval time by 35% by designing and implementing scalable data processing pipelines for application logs and system metrics on AWS, covering end-to-end ingestion, transformation, storage, and monitoring of operational data.

IoT Intern 05/2023 – 08/2023
Parity Infotech Solutions New Delhi, India
- Improved real-time data processing efficiency by 20% by developing sensor and IoT traffic data integration pipelines in Python, supporting smart city infrastructure with reliable data flows.
- Restored software stability and reduced recurring failures by diagnosing and resolving 20+ performance-critical defects through systematic analysis of system logs, and metrics.
- Accelerated team debugging capability by documenting defect patterns, root causes, and resolution steps, creating a reusable reference that reduced investigation time for similar issues.

Skills

Languages & Backend: Python, REST APIs, Microservices, System Design, Data Structures & Algorithms

Cloud, DevOps & Automation: AWS (EC2, S3, RDS, VPC, Lambda, CloudFront), Terraform, Docker, Kubernetes, GitHub Actions, GitLab CI/CD, Jenkins

Observability, Data & AI: Prometheus, Grafana, OpenTelemetry, Log Analysis, Data Pipelines, Anomaly Detection, PostgreSQL

OS & Tools: Linux (CLI, process management, log reading), Bash, Git, GitHub, GitLab, On-premise Deployment, Microsoft 365, Tableau

Projects

AWS Django Blog Application
Blog Application AWS (EC2, RDS, S3, ALB, CloudFront, Lambda, DynamoDB) · Docker · Python · Django · MySQL · CI/CD
- Improved system responsiveness under peak load by 30% by implementing auto-scaling groups, Application Load Balancer, and CloudFront CDN distribution across a production-grade AWS deployment.
- Achieved high availability and fault tolerance for a cloud-native application by architecting VPC isolation, RDS MySQL database integration, and Route 53 DNS failover for zero-downtime releases.
- Reduced deployment time and manual effort by building automated CI/CD pipelines with Docker containerization and infrastructure-as-code practices, enabling reproducible and consistent environment provisioning.

Python Automation Tool - Clean My Downloads
Python · CLI · Cross-platform (macOS, Linux, Windows) · Automation
- Reduced manual file management effort by 80% by developing a cross-platform Python CLI tool that automatically classifies and organizes files across macOS, Linux, and Windows environments.
- Ensured broad compatibility and zero external dependencies by building the tool entirely on Python's standard library, making it lightweight and portable across all major operating systems.
- Improved maintainability and usability by implementing modular code architecture and clear documentation, enabling straightforward extension of file classification rules and CLI usage by other developers.

AI-Based Log & Sensor Data Processing Pipeline (Proof of Concept)
- mproved data processing efficiency by 25% by designing and benchmarking end-to-end ingestion, transformation, and storage pipelines for large-scale structured and unstructured log and sensor data.
- Reduced manual monitoring effort by integrating Generative AI models to automatically summarize logs, detect anomalies, and generate actionable insights for system observability.
- Enabled scalable and reproducible experimentation by containerizing the full application stack with Docker and deploying on AWS, supporting consistent execution across development and cloud environments.

Certifications and Awards

Ultimate AWS certified Cloud Practitioner | Ultimate AWS Certified Solutions Architect Associate | IEEE MTT-S Winner

Languages

English (C2) · German (A1)